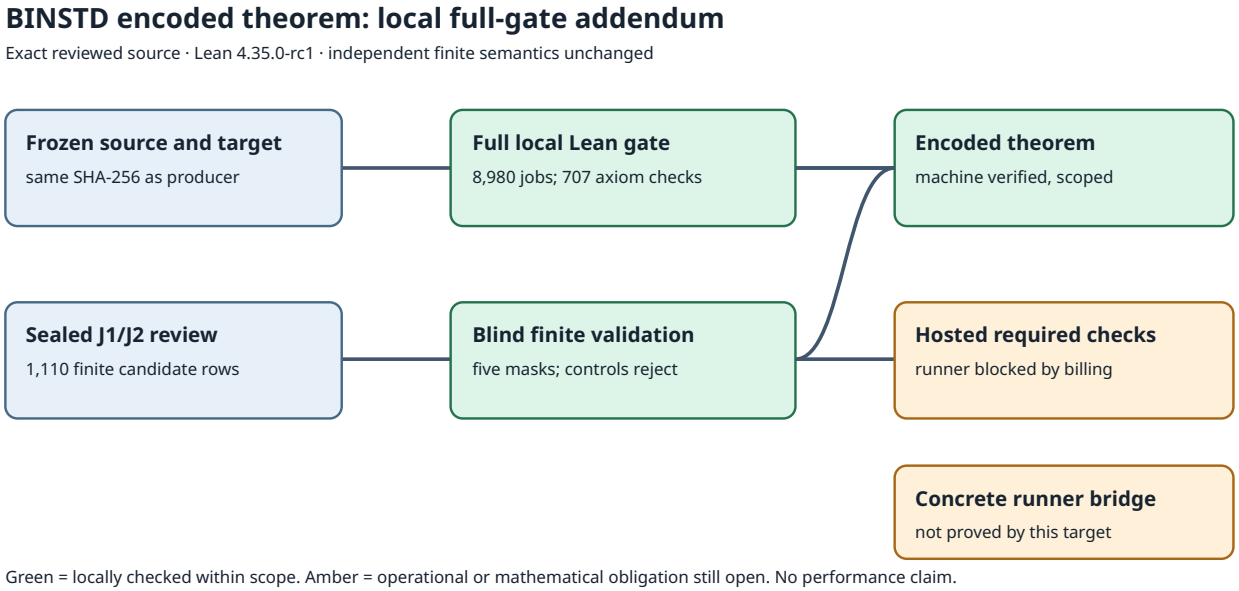
BINSTD encoded theorem: local full-gate addendum

27 October 2026 UTC Frozen producer `6409590d` Execution head `98fca453`



Observation. The pinned Lean 4.35.0-rc1 root built 8,980 Lake jobs. The repository's exact formal verifier returned `machine_verified`: the named module built, the project audit found 707 theorem declarations using only permitted standard axioms, and the requested-theorem audit passed. The source, target, toolchain and manifest hashes match the independently reviewed producer inputs. The verifier found no forbidden construct or infrastructure failure.

Review claim	Sealed snapshot	2026-10-27 local update
C1 finite masks/counts	validated scoped	unchanged: 1,110 rows independently retained
C2 Lean declaration	direct check only	full exact-target gate passed for identical source
C3 direct receipt	validated direct scope	unchanged
C4 full workspace	not verified	locally machine verified; hosted job did not start
C5 polynomial/runner bridge	open	still open for this encoded theorem

Boundary. The theorem checks the specified finite XOR-and-shift predicate. It does not connect that function to polynomial Euclidean remainder or the archived runner's Frobenius representation, and it supplies no yield or attack-cost measurement. Separate PR #2484 contains unmerged cyclotomic and abstract finite-field proofs. Current-head GitHub Actions are blocked before runner assignment by the account payment/spending-limit gate; merge and Pages publication remain pending.

Evidence. This write-once addendum cites the sealed J1/J2 packet and the new SHA-bound full-gate validation receipt in the adjacent BINSTD study. The original review archive remains unchanged. No official scientific status transition is made.